

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Mixed

10 questions · ~15 min

01

Q1

GEOMETRY AND TRIGONOMETRY

The table gives the perimeters of similar triangles TUV and XYZ , where TU corresponds to XY . The length of TU is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of XY ?

- A. 2
- B. 18
- C. 55
- D. 162

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

The length of each edge of a box is 29 inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A certain elephant weighs 200 pounds at birth and gains more than 2 but less than 3 pounds per day during its first year. Which of the following inequalities represents all possible weights w , in pounds, for the elephant 365 days after birth?

- A. $400 < w < 600$
- B. $565 < w < 930$
- C. $730 < w < 1,095$
- D. $930 < w < 1,295$

A number x is at most 17 less than 5 times the value of y . If the value of y is 3, what is the greatest possible value of x ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = 2x + 10$$

$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the xy -plane?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

$$(5x^3 - 3) - (-4x^3 + 8)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN ADVANCED MATH

$$y = 4x$$

$$y = x^2 - 12$$

A solution to the given system of equations is x, y , where $x > 0$. What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = x - 2x + 4$$

$$y = 6x - 12$$

Which ordered pair x, y is the solution to the given system of equations?

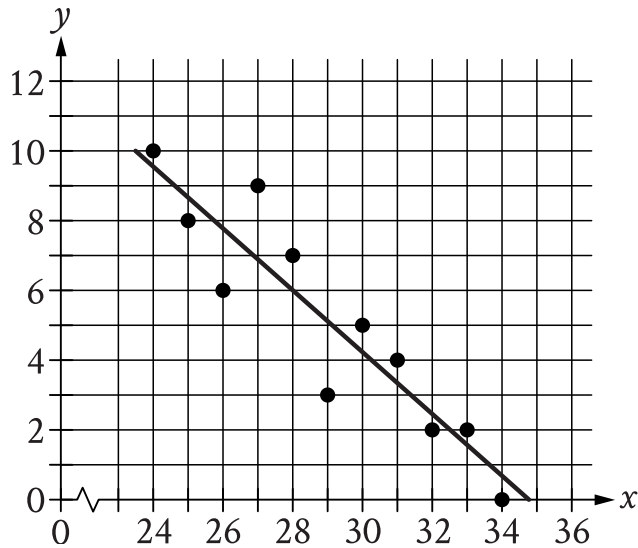
- A. 0, 2
- B. -4, 2
- C. 2, 0
- D. 2, -4

$$d = 55t$$

The equation above can be used to calculate the distance d , in miles, traveled by a car moving at a speed of 55 miles per hour over a period of t hours. For any positive constant k , the distance the car would have traveled after $9k$ hours is how many times the distance the car would have traveled after $3k$ hours?

- A. 3
- B. 6
- C. $3k$
- D. $6k$

The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



- The scatterplot has 11 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
 - The line of best fit slants down from left to right.
 - 6 points are above the line of best fit.
 - 5 points are below the line of best fit.
 - The line of best fit goes through the following approximate coordinates:
 - (28 comma 6)
 - (33 comma 1.5)

At $x = 32$, which of the following is closest to the y -value predicted by the line of best fit?

A. 0.4

B. 1.5

C. 2.4

D. 3.3